



GRACE COLLEGE OF ENGINEERING
(Approved by AICTE, New Delhi & Affiliated to ANNA University, Chennai)
MULLAKKADU, THOOTHUKUDI - 628 005

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

B.Tech- Artificial Intelligence And Data Science

Anna University Regulation: 2021

AL3391- Artificial Intelligence

II Year/III Semester

Unit- IV Logical Reasoning

Prepared By,

Mrs. R. Janani, AP/AI&DS

Unit- IV Logical Reasoning

PART A

1. Define FOL.

FOL is a first order logic. It is a representational language of knowledge which is powerful than propositional logic (i.e.) Boolean Logic. It is an expressive, declarative, compositional language.

2. Define a knowledge Base

Knowledge base is the central component of knowledge base agent and it is described as a set of representations of facts about the world.

3. With an example, show objects, properties functions and relations.

Example “EVIL KING JOHN BROTHER OF RICHARD RULED ENGLAND IN 1200” Objects : John, Richard, England, 1200 Relation : Ruled Properties : Evil, King Functions : BROTHER OF

4. Define a Sentence?

Each individual representation of facts is called a sentence. The sentences are expressed in a language called as knowledge representation language.

5. Define an inference procedure

An inference procedure reports whether or not a sentence is entailed by knowledge base provided a knowledge base and a sentence. An inference procedure ‘i’ can be described by the sentences that it can derive. If i can derive from knowledge base, we can write. KB $\rightarrow$ Alpha is derived from KB or i derives alpha from KB.

6. What are the three levels in describing knowledge based agent?

• Logical level • Implementation level • Knowledge level or epistemological level

7. Define Syntax? Syntax is the arrangement of words. Syntax of a knowledge describes the possible configurations that can constitute sentences. Syntax of the language describes how to make sentences.

8. Define Semantics

The semantics of the language defines the truth of each sentence with respect to each possible world. With this semantics, when a particular configuration exists within an agent, the agent believes the corresponding sentence.

9. Define Logic Logic is one which consists of i. A formal system for describing states of affairs, consisting of a) Syntax b) Semantics. ii. Proof Theory – a set of rules for deducing the entailment of a set of sentences.

10. What are the 3 types of symbols which are used to indicate objects, relations and functions?

- i) Constant symbols for objects ii) Predicate symbols for relations iii) Function symbols for functions

11. Define terms.

A term is a logical expression that refers to an object. We use 3 symbols to build a term.

12. Define Atomic sentence.

Atomic sentence is formed by both objects and relations.

Example Brother (William, Richard) William is the brother of Richard.

13. Define Quantifier and it's types.

Quantifiers are used to express properties of entire collection of objects rather than representing the objects by name.

Types: i. Universal Quantifier ii. Existential Quantifier iii. Nested Quantifier.

14. Define a Proof

A sequence of application of inference rules is called a proof. Finding proof is exactly finding solution to search problems. If the successor function is defined to generate all possible applications of inference rules then the search algorithms can be applied to find proofs.

15. Define a Complete inference procedure An inference procedure is complete if it can derive all true conditions from a set of premises.

16. Define Interpretation

Interpretation specifies exactly which objects, relations and functions are referred to by the constant predicate, and function symbols.

17. Define Validity of a sentence

A sentence is valid or necessarily true if and only if it is true under all possible interpretation in all possible world.

18. Define Satisfiability of a sentence

A sentence is satisfiable if and only if there is some interpretation in some world for which it is true

19. What are the basic Components of propositional logic?

- i. Logical Constants (True, False)

20. Define Modus Ponens's rule in Propositional logic?

The standard patterns of inference that can be applied to derive chains of conclusions that lead to the desired goal is said to be Modus Ponens's rule.

21 .Define AND –Elimination rule in propositional logic

AND elimination rule states that from a given conjunction it is possible to inference any of the conjuncts.
OR-Introduction rule states that from, a sentence, we can infer its disjunction with anything.

22. Define Unification.

Lifted Inference rule require finding substitutions that make different logical expressions look identical (same). This is called Unification.

23. Define backward chaining.

This algorithm works backward from the goal, chaining through rules to find known facts that support the proof.

24. Define Prolog program.

It is a set of definite clauses written in a notation somewhat different from standard FOL.

25. State the converse and contrapositive of the statement "When I stay up late, it is necessary that I sleep until noon".

- **Converse:** "When I sleep until noon, it is necessary that I stayed up late."
- **Contrapositive:** "When I do not sleep until noon, it is necessary that I did not stay up late"

26. Define the terms belief state and state estimation

- **Belief State:** The knowledge or assumptions an agent has about the current state of the environment, often represented probabilistically to account for uncertainty.
- **State Estimation:** The process of determining the most likely current state of a system based on observations and a model of the system, reducing uncertainty and filtering out noise from the data.

27. What are the three levels in describing a knowledge-based agent?

- **Knowledge Level:** This level describes the beliefs, goals, and knowledge of the agent. It is the "what" level, focusing on what the agent knows and aims to achieve.
- **Logical Level:** This level details how the agent's knowledge is represented and manipulated. It involves formal logical systems, rules, and inference mechanisms that the agent uses to reason.
- **Implementation Level:** This level covers the actual implementation of the agent, including the data structures and algorithms used to realize the knowledge and logical levels in a working system.

27. State Unification in first order logic

Unification in first-order logic is a process that determines if two logical expressions can be made identical by substituting variables with terms. It's a key concept in automated reasoning, allowing us to match patterns and derive logical conclusions.

The unification algorithm systematically finds the most general unifier (MGU), which is the simplest substitution that makes the expressions identical. Unification is essential in logic programming and theorem proving, such as in Prolog and automated theorem provers.

28. If the unicorn is mythical, then it is immortal, but if it is not mythical, it is a mortal mammal. If the unicorn is either immortal or a mammal, then it is horned. The unicorn is magical if it is horned. Prove: the unicorn is horned.

29. Some people like every vegetable. Convert it to First Order Logic.

PART-B

- 1.Explain the Propositional logic with example.
- 2.Discuss Forward and Backward chaining
- 3.Discuss the syntax and semantics of FOL.
- 4.Explain the Resolution procedure.
- 5.Explain the Steps involved in knowledge engineering process.
- 6.Discuss the algorithm for unification.
- 7.Differentiate FOL and Propositional logic(Prolog).

Express the following statements as predicates:

- All people who are not poor and are smart are happy.
- Those people who read are not stupid.
- John can read and is wealthy.
- Happy people have exciting lives.

Skolemize the above statements if required and prove the following by resolution: "Can anyone be found with an exciting life?" (Assume stupid = not smart, wealthy = not poor).

8.During a murder investigation, you have gathered some clues. Express them as propositions and solve the following scenario using laws of inference. The clues gathered are given below:

- If the knife is in the store room, then we saw it when we cleared the store room.
- The murder was committed at the basement or inside the apartment.
- If the murder was committed at the basement, then the knife is in the yellow dust bin.
- We did not see a knife when we cleared the store room.
- If the murder was committed outside the building, then we are unable to find the knife.

- If the murder was committed inside the apartment, then the knife is in the store room.

Find: "Where is the knife?"

9. (i) What are logical connectives? Explain in detail.

(ii) Describe an algorithm for general propositional inference based on model checking

10. a. Discuss the Knowledge Engineering Process with proper illustration.

b. Depict the concept of forward chaining.

11. a. Brief on the concept of Resolution and explain the Propositional Resolution Algorithm.

b. Prove the following axioms using the Resolution Algorithm:

1. All hounds howl at night.
2. Anyone who has any cats will not have any mice.
3. Light sleepers do not have anything which howls at night.
4. John has either a cat or a hound.
5. (Conclusion) If John is a light sleeper, then John does not have any mice.

12. Give the rules of inferences in Propositional Logic. (ii) Which rule of inference is used in each argument below?

1. Alice is a Math major. Therefore, Alice is either a Math major or a CSI major.
2. Jerry is a Math major and a CSI major. Therefore, Jerry is a Math major.
3. If it is rainy, then the pool will be closed. It is rainy. Therefore, the pool is closed.
4. If it snows today, the university will close. The university is not closed today. Therefore, it did not snow today.
5. If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will sunburn. Therefore, if I go swimming, then I will sunburn.

13. The missionaries and cannibals problem is as follows: Three missionaries and three cannibals are on one side of a river, along with a boat that can hold one or two people. Find a way to get everyone to the other side without ever leaving a group of missionaries in one place outnumbered by the cannibals in that place.

- (i) Formulate the problem precisely, making only those distinctions necessary to ensure a valid solution. Draw a diagram of the complete state space.
- (ii) Implement and solve the problem optimally using an appropriate search algorithm. Is it a good idea to check for repeated states?
- (iii) Why do you think people have a hard time solving this puzzle, given that the state space is so simple?